Annotation of novel miRNAs in *Brassicaceae oleracea*

Shel Burkes

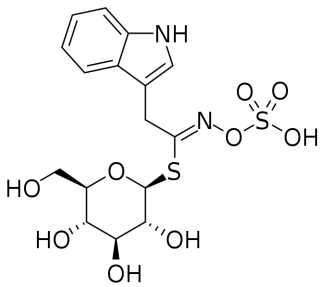
Brassicaceae oleracea var. italica

- The word broccoli comes from the Italian plural of broccolo, which means "the flowering crest of a cabbage"[8].
- Cultivars in the family all possess mutations that lead to a variety of phenotypes/genotypes observed.
  - The most well known variety of *oleracea* is Calabrese broccoli.

The most well known variety of oleracea is Calabrese broccoli. [source]

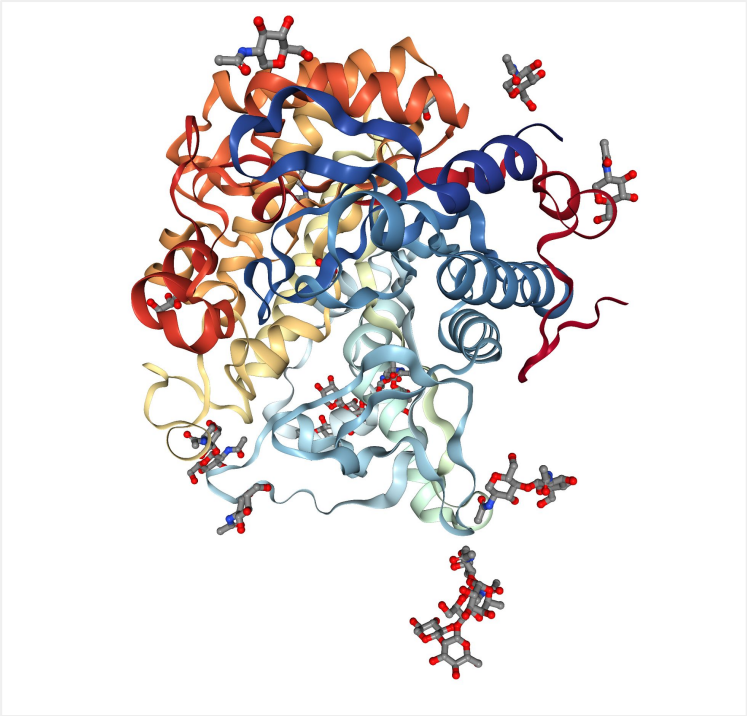
Phytochemicals of importance: Glucosinolates

- Glucosinolates are sulfur-rich secondary metabolites that occur in order brassicales.
- Organic compounds that contain sulfur and nitrogen and are derived from glucose and an amino acid.
- 120 side chain structures for glucosinolates have been described[4], only a restricted number of side chain structures occur in commonly consumed cruciferous crops [3].
- Glucosinolates of importance include:
  - Glucoraphanin -> Sulforaphane (SFN) - *Breast Cancer*
  - Glucobrassicin -> Indole-3-carbinol (I3C) - *Ulcerative Colitis*



Glucosinolate Pathway

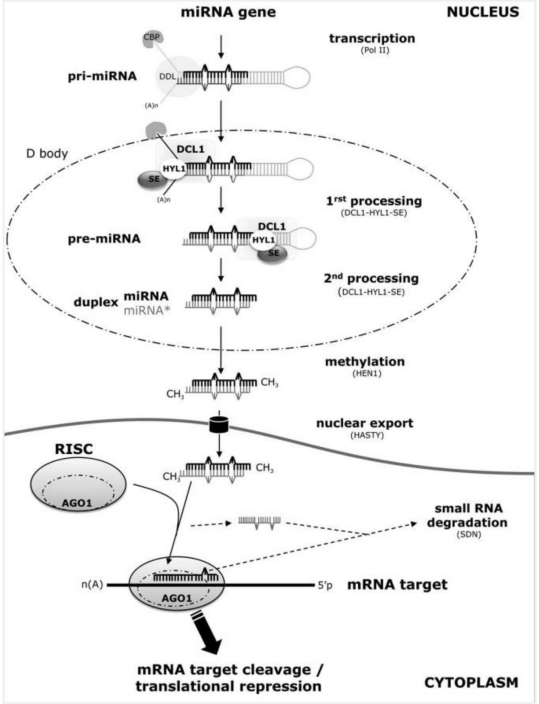
- There are 3 primary variants of the pathway:
  - Branched Chain Amino Acids
  - Aromatic Amino Acids
  - Methionine
- Glucosinolates are subsequently broken down by myrosinase.



Myrosinase from Sinapis alba (White Mustard) [source]

miRNA

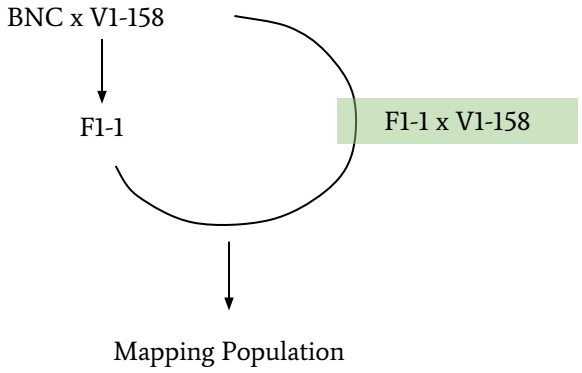
- Small non-coding RNA molecule (containing approximately 22 nucleotides) found in plants, animals and some viruses.
- Functions in RNA silencing and post-transcriptional regulation of gene expression.
- Plant miRNA biogenesis slightly differentiated from animal pathway
  - Dicer-like protein 1 vs. Dicer



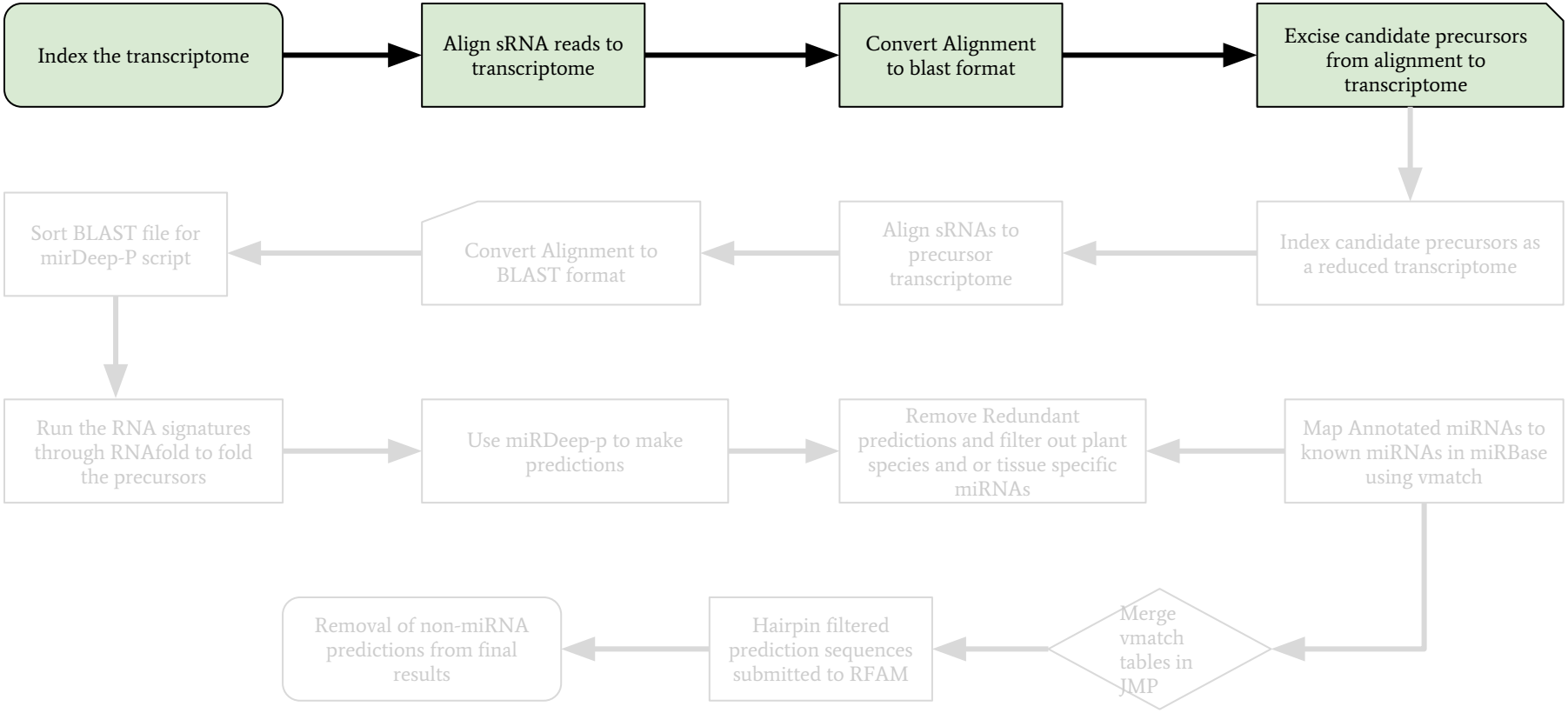
miRNA Biogenesis in Plants [12]

Project Goals

1. Annotate broccoli miRNAs in parents of a mapping population
2. Identify candidate miRNAs that control targeted health impacts

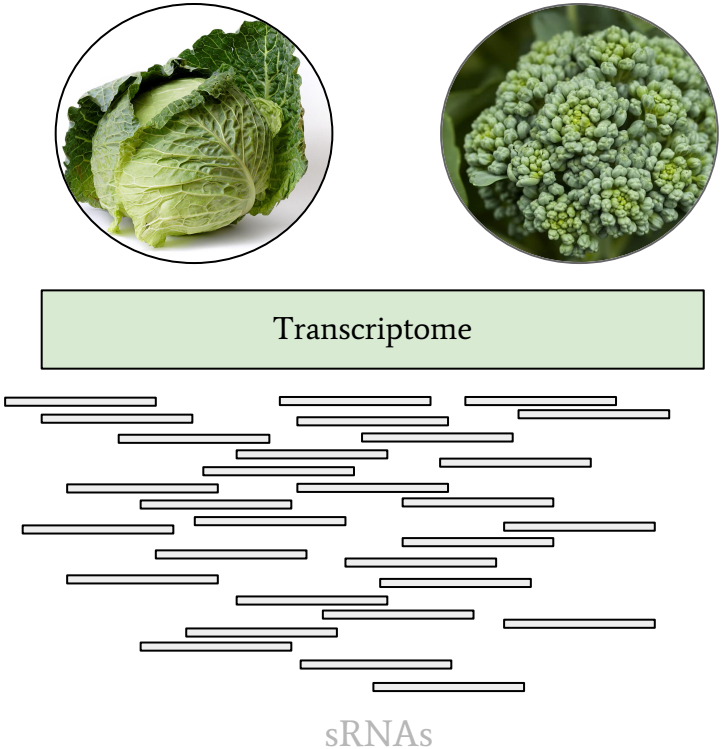


Workflow


Index the transcriptome
Align sRNA reads to transcriptome
Convert Alignment to blast format
Excise candidate precursors from alignment to transcriptome
Index candidate precursors as a reduced transcriptome
Align sRNAs to precursor transcriptome
Convert Alignment to BLAST format
Sort BLAST file for mirDeep-P script
Run the RNA signatures through RNAfold to fold the precursors
Use miRDeep-p to make predictions
Remove Redundant predictions and filter out plant species and or tissue specific miRNAs
Map Annotated miRNAs to known miRNAs in miRBase using vmatch
Merge vmatch tables in JMP
Hairpin filtered prediction sequences submitted to RFAM
Removal of non-miRNA predictions from final results

Transcriptome Indexing & Alignment

- Transcriptome was indexed using Bowtie
  - Transcriptome was assembled by Dr. Reid at the DHMRI in Kannapolis, NC
- Pre-processed sRNAs formatted for miRDeep-p were then aligned.
- Alignment converted to BLAST format.



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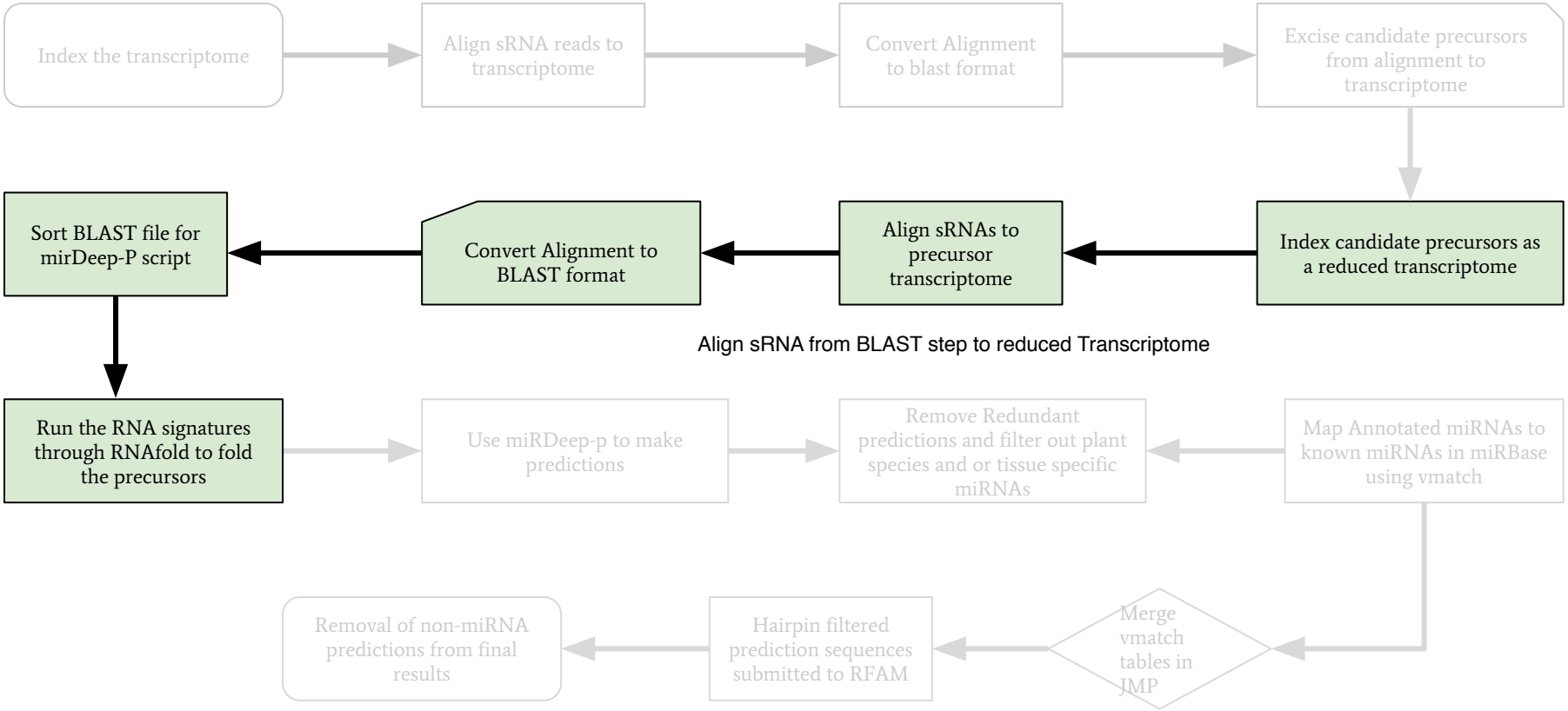
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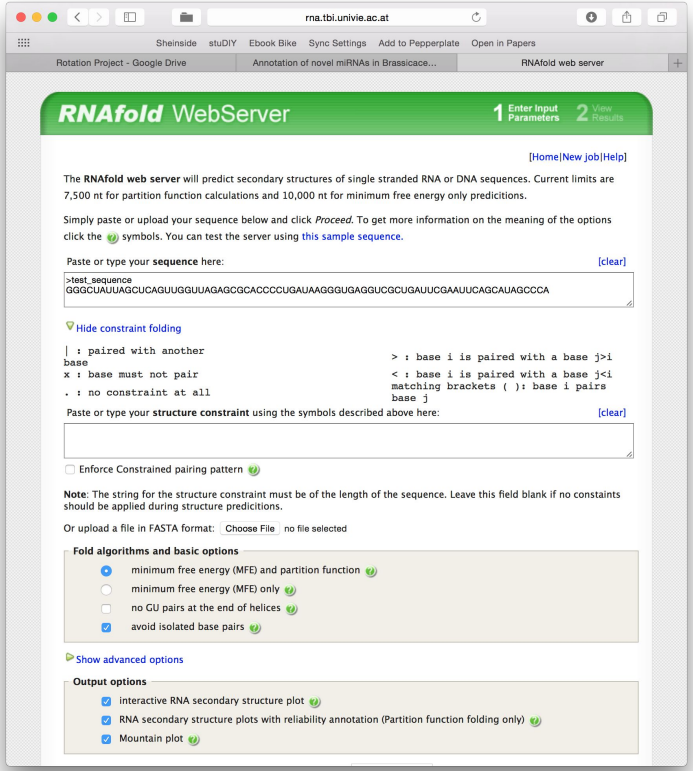
Candidate Precursors

- Excision of candidate precursors from alignment.
  - 276 was the length for excision.
- Candidate precursors were used a reduced size transcriptome, and indexed again using bowtie.
- Alignment was converted to BLAST format.
  - BLAST format is the necessary file input for miRDeep-p

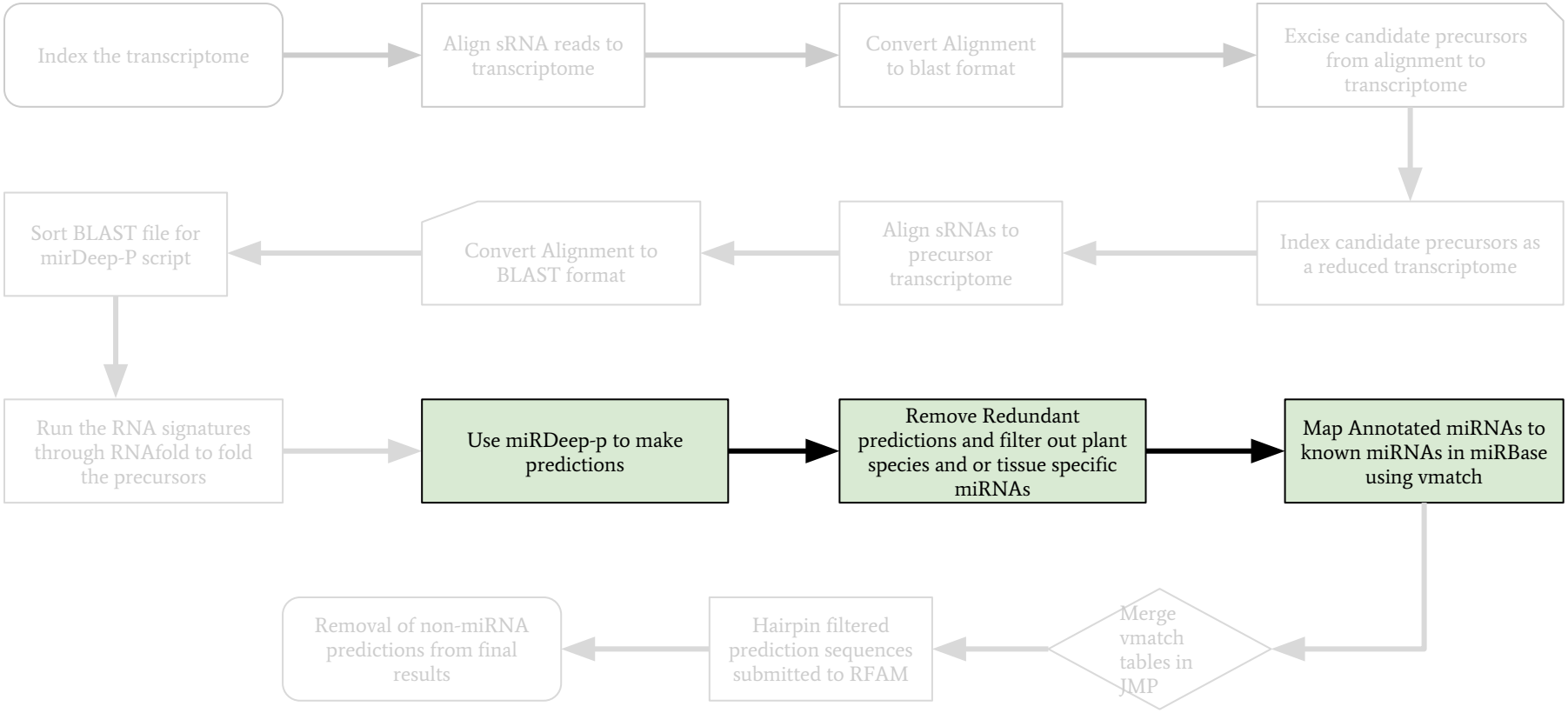
RNAfold - Precursor folding

- Predicts secondary structures of single stranded RNA or DNA sequences.
- Computed using RNAfold (Vienna)
- Possible inputs:
  1. Fasta file with deep sequencing reads.
  2. Fasta file of the corresponding genome.
  3. File of mapped reads to the genome in miRDeep2 arf format.
  4. Fasta file with known miRNAs of the analysing species.
  5. An option fasta file of known miRNAs of related species.
- Precursors that fold into the traditional hairpin-loop structure will be output into a .fasta file.



WebServer RNAFold [source]

Workflow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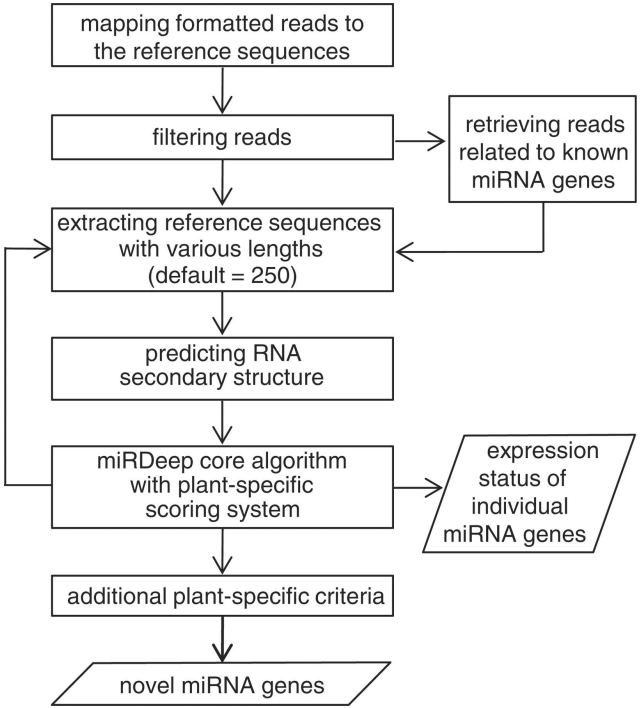
miRDeep-p

- Output from folding predictions are then run with miRDeep to identify novel miRNAs.
  - Uses folding structures and precursors to detect novel miRNA genes
- The script, *Rm_redundancy_meet_plant.pl,* removed redundant items and filtered the items out that do not meet the criteria of plant miRNAs.
  - Criteria set by sRNA precursors predictions, sRNA non-redundant predictions, and sRNA filtered predictions.
  - F1-1: 33 hairpin precursors representing 19 unique miRNAs
  - VI-158: 59 hairpin precursors representing 32 unique miRNAs



Flowchart of miRDeep workflow[8]

vMatch

- Preprocesses a set of sequences into an index structure.
- This is stored as a collection of several files constituting the persistent index, somewhat similar to Bowtie.

Analysis Workflow:

- Filtered sRNA sequences from miRDeep prediction were made into a fasta.
- Indexed miRBase using vmatch.
- Predictions aligned to miRBase using vmatch (miRBase v21 - mature.fa).
    - Used direct matching, with a seed of 18 and a mismatch of 3 bases between the subject and query sequences.

Conserved miRNA Families

Conserved miRNA Family	Target Transcription Factor Family	Function
miR156	SBP (SQUAMOSA PROMOTER BINDING PROTEIN)	Vegetative growth
miR159/miR319	MYB, TCP	Anther and floral development
miR160	ARF (AUXIN RESPONSE FACTOR)	Hormone signal transduction
miR164	NAM/CUC	Floral and shoot development
miR165/miR166	HD-ZIP	Vascular development, meristem formation
miR167	ARF (AUXIN RESPONSE FACTOR)	Hormone signal transduction
miR170/miR171	SCL (SCARECROW-LIKE)	Flower development
miR172	AP2 (APETALA2)	Flower development
miR396	GRF (GROWTH REGULATING FACTOR)	Growth and development

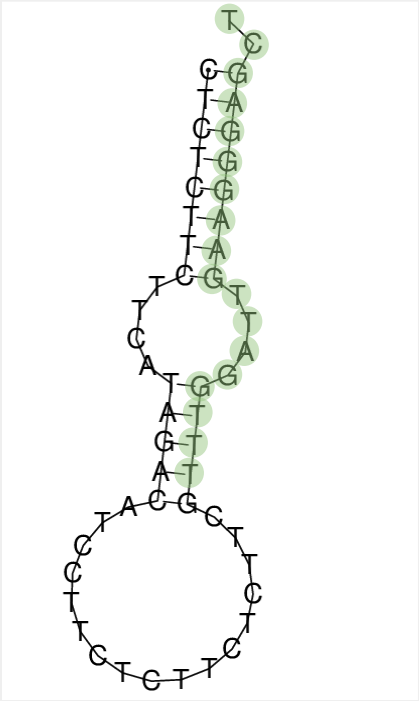
miR159

- Identified in vMatch results
  - Arabidopsis thaliana has annotation of 3 miR159 family members
  - Brassica rapa and brassica napa have 1 miR159 annotated in miRBase
  - Two different transcripts in VI-158 produces two identical hairpins.

- Read Counts
  - In both parental cases, this indicates transcriptome assembly *does not* cover enough hairpin sequence or that the sequence failed plant-specific criteria.

	Sequence	Count
miRBase	TTTGGATTGAAGGGAGCTCTA	
*VI-158	TTTGGATTGAAGGGAGCT	7
VI-158	TTTGGATTGAAGGGAGCTCTA	2406
*F1-1	TTTGGATTGAAGGGAGCTCTA	942

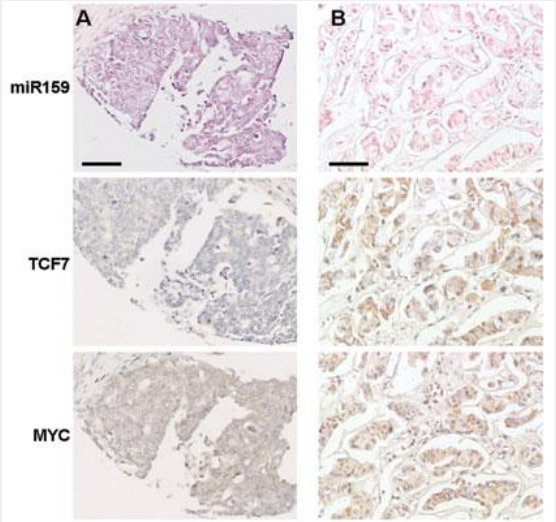
* not predicted by miRDeep-p but present in small RNA library



RNAfold generated image of precursor miR159. Areas of green denote the miRNA sequence.

Implications in Human Cancer: miR159 mimics

- A miR159 mimic has been shown to inhibit cancer proliferation by targeting TCF7 gene that encodes for a Wnt signaling transcription factor [11] that is upregulated in breast cancer.
    - The Wnt pathway is an evolutionarily conserved pathway that regulates crucial aspects of cell fate determination, cell migration, cell polarity, neural patterning and organogenesis during embryonic development.[13]
- In the study, it was found that miR159 levels were inversely correlated with levels of TCF7 and MYC in breast tumors.



Correlation	miR159 vs. TCF7	miR159 vs. MYC	TCF7 vs. MYC
Pearson *r* value	-0.266	-0.224	0.543
p value	0.013	0.039	<0.001
n	86	86	86

Future Work

- MiRNA annotation using the newest release of the broccoli genome for better coverage.
- Incorporation of Metabolite Data
  - Differential expression of miRNAs across mapping population
  - Correlate miRNA expression to metabolite data (LC-MS and NMR)
- Predictive Modelling
  - Prediction of genetic markers, generating a model, in a larger population.
  - Traits of interest
  - Target prediction: http://plantgrn.noble.org/psRNATarget/

Acknowledgments

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DHMRI

- Dr. Rob Reid

NCSU

- Dr. Alan Brown

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Thank You!